quantum mechanics from scratch, in which measurements do have definite outcomes, Bell's inequality is the most important guidepost you have to keep in mind. If, on the other hand, you're happy with Many-Worlds and are trying to puzzle out how to map the theory onto our observed experience, Bell's result is an automatic consequence of the underlying equations, not an additional constraint you need to worry about moving forward.

One of the fantastic things about Bell's theorem is that it turns the supposed spookiness of quantum entanglement into a straightforwardly experimental question—does nature exhibit intrinsically non-local correlations between faraway particles, or not? You'll be happy to hear that experiments have been done, and the predictions of quantum mechanics have been spectacularly verified every time. There is a tradition in popular media of writing articles with breathless headlines like "Quantum Reality Is Even More Bizarre Than Previously Believed!" But when you look into the results they are actually reporting, it's another experiment that confirms exactly what a competent quantum mechanic would have predicted all along using the theory that had been established by 1927, or at least by 1935. We understand quantum mechanics enormously better now than we did back then, but the theory itself hasn't changed.

Which isn't to say that the experiments aren't important or impressive; they are. The problem with testing Bell's predictions, for example, is that you are trying to make sure that the extra correlations predicted by quantum mechanics couldn't have arisen due to some sneaky pre-existing classical correlation. How do we know whether some hidden event in the past secretly affected how we chose to measure our spin, or what the measurement outcome was, or both?

Physicists have gone to great lengths to eliminate these possibilities, and a cottage industry has arisen in doing "loophole-free Bell tests." One recent result wanted to eliminate the possibility that an unknown process in the laboratory worked to influence the choice of how to measure the spin. So instead of letting a lab assistant choose the measurement, or even using a random-number generator sitting on a nearby table, the experiment made that choice based on the polarization of photons emitted from stars many light-years away. If there were some nefarious conspiracy to make the world look quantum-mechanical, it had to have been set up hundreds of